

The Synthesis of a New Family of Boron-Based Anion Receptors and the Study of Their Effect on Ion Pair Dissociation and Conductivity of Lithium Salts in Nonaqueous Solutions

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ABSTRACT

A new family of anion receptors based on boron compounds has been synthesized. These compounds can be used as anion receptors in lithium battery electrolytes. This family includes various borane and borate compounds with different fluorinated aryl and fluorinated alkyl groups. When these anion receptors are used as additives in 1,2-dimethoxyethane (DME) solutions containing various lithium salts, the ionic conductivities of these solutions are greatly increased. The electrolytes tested in this study were DME solutions containing the following lithium salts: LiF, LiCl, LiBr, LiI, CF_3COOLi , and $\text{C}_2\text{F}_5\text{COOLi}$. Without the additive, the solubility of LiF in DME (and all other nonaqueous solvents) is very low. With some of these boron compounds as additives, LiF solutions in DME with concentration as high as 1 M were obtained. The solubilities of the other salts were also increased by these additives. Near edge X-ray absorption fine structure (NEXAFS) spectroscopy studies show that Cl^- and I^- anions are complexed with these compounds in DME solutions containing LiCl or LiI salts. The degree of complexation is also closely related to the structures of the fluorinated aryl and alkyl groups, which act as electron-withdrawing groups. The NEXAFS results are in good agreement with ionic conductivity studies.

Introduction

Reduction of ion pairing in nonaqueous electrolytes for lithium batteries is an important way for improving the properties of these electrolytes. There are two beneficial results from ion-pairing reduction. One is to increase the ionic conductivity of the electrolytes by increasing the number of charge carriers; the other is enhancement of the lithium ion transference number. The effects on the lithium ion transference number by ion pairing have been examined by Doyle et al.¹ using computer modeling. Currently, the ion-pair reduction is achieved by using a solvent with a high dielectric constant and salts that are easily dissociated. A third approach is to develop new additives to dissociate the ion pairs. The dissociation is accomplished by forming complexes of either cations or anions with the additives. Cation receptors have been used as additives in lithium battery electrolytes in several studies. Saloman² has reported on the use of the crown ether 18-crown-6, and Matsuda³ on the use of 12-crown-4, for cation coordination in liquid nonaqueous electrolytes. Lonergan et al.⁴ have similarly used cryptands in polymer electrolytes. Anion complexation has not been widely studied in this field because of a lack of suitable anion receptors. However, anion coordination is much more important than cation coordination in lithium battery electrolytes since it offers the promise of both increasing the conductivity and the lithium ion transference number. Thus the achievement of anion coordination could open a new approach for lithium battery electrolyte development.

The design and synthesis of receptor molecules for the selective complexation of ions has been an active area for the past two decades. Most of the work is in the biological area with the emphasis on molecular recognition and the mimicking of processes in biological membranes. Still the number of host molecules for anions is very small. To accomplish anion binding, most of the reported host molecules contain either positively charged sites or Lewis acid metal centers which restricted their application in lithium battery electrolytes. In the last two years, there have been a few reports on the use of neutral compounds to complex anions. These compounds were summarized in a review article by Schmidtchen and Berger.⁵ However, anion com-

plexation in these compounds is based on hydrogen bonding, and such compounds would not be stable in lithium batteries.

Previously, we reported on the synthesis of a family of aza-ether based compounds as anion receptor molecules which are not based on either positively charged sites or hydrogen bonding. An electron-withdrawing group (R), $\text{R}=\text{CF}_3\text{SO}_2$, was used to substitute amine hydrogen atoms in aza-ethers.⁶ In this paper, we report the synthesis of another family of anion receptors based on boron compounds. These compounds could be used as additives in nonaqueous electrolytes for lithium batteries. The details of the synthesis and structure of these compounds are discussed. This paper also reports on the effect of these agents as additives on the conductivity of LiCl/DME, LiF/DME, LiI/DME, CF_3COOLi /DME, and $\text{C}_2\text{F}_5\text{COOLi}$ /DME electrolytes. Compared to the lithium salts used for commercial lithium battery electrolytes, such as LiPF_6 and LiAsF_6 , these salts are low cost and less toxic. However, these advantages cannot be utilized due to their low conductivities in nonaqueous electrolytes. If the ionic conductivities of these electrolytes could be improved by using boron-based anion receptors as additives, it would offer the possibility of replacing LiPF_6 or LiAsF_6 by these salts. The relationship between the conductivity enhancement and the structure of the anion receptors is discussed. Near edge X-ray absorption fine structure (NEXAFS) spectroscopy at the Cl K edge and the I L_{α} edge was used to study the complexation between these anion receptors and the Cl^- or I^- anions.

Experimental

Synthesis and characterization.—The structures of the synthesized compounds and the reference compounds are given in Fig. 1. The number of each structure corresponds to the compound number in the text. Compound 1 and compound 7 are commercially available and were used as reference compounds in this study. The other 12 compounds in Fig. 1 are new. The synthesis methods and nuclear magnetic resonance (NMR) data for the new compounds are given below. The starting compounds were purchased from Aldrich Chemical Co. All the reactions and processes were performed under an argon atmosphere or under vacuum. NMR spectra were recorded on a Hitachi R-1200 (60 MHz) NMR spectrometer.

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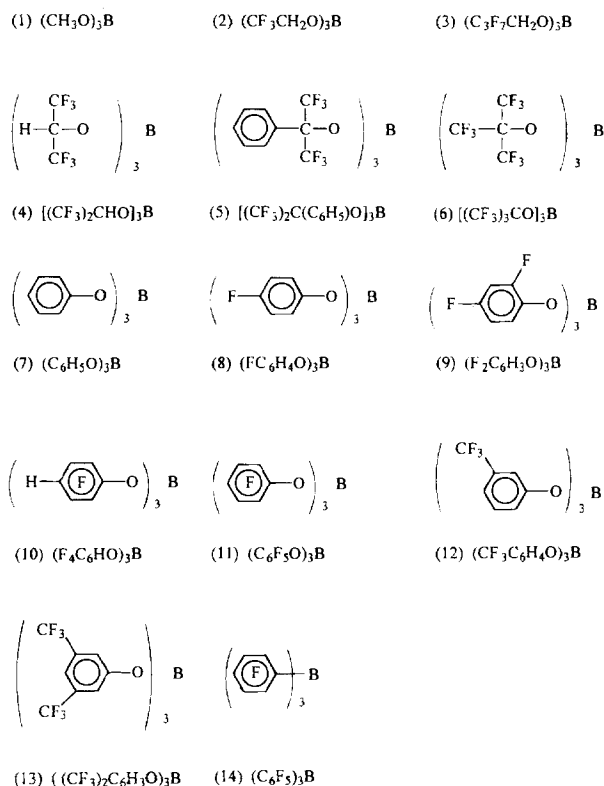


Fig. 1. Chemical structure of boron-based anion receptors and related reference compounds. The numbers in parentheses correspond to the compound number in the text.

General procedure for preparation of the fluorinated alkyl borate compounds: compound 2, 3, 4, and 5.—0.12 mol of the corresponding fluorinated alkyl alcohol were added dropwise over a period of 15 min, to 4 mL of borane-methyl sulfide complex (total 0.04 mol) in a cold bath at -20°C . The mixture was then stirred at room temperature for 12 h. 100 mL of pentane was added to the mixture. After removal of all low boiling point materials (methyl sulfide and pentane) through distillation, the final products were further purified through either sublimation or distillation under vacuum. The boiling temperatures and respective NMR data were as follows

Compound 2: tris(2,2,2-trifluoroethyl) borate, boiling point: 78 to $80^\circ\text{C}/100\tau$; NMR (in CDCl_3): 4.2(q). (unit: ppm).

Compound 3: tris(1H,1H-heptafluorobutyl) borate, boiling point: 95 to $96^\circ\text{C}/16\tau$; NMR (in CDCl_3): 4.35(t). (unit: ppm).

Compound 4: tris(2H-hexafluoroisopropyl) borate, boiling point: 78 to $80^\circ\text{C}/100\tau$; NMR (in CDCl_3): 5.8(q). (unit: ppm).

Compound 5: tris(perfluoro-*t*-butyl) borate, boiling point: $52^\circ\text{C}/7.5\tau$.

Procedure for preparation of compound 6.—A solution of 9.8 g hexafluoro-2-phenylisopropanol in 10 mL ether was added dropwise to a mixture of 1.06 g sodium hydride and 100 mL anhydrous ether in a cold bath at -20°C . Then the solution was warmed up to room temperature. After the reaction was completed, the solution was cooled to -40°C . 13.3 mL of 1 M boron trichloride in hexane was added to the solution over a period of 20 min. The solution was stirred at -40°C for 1 h, then warmed to room temperature and stirred for another 12 h. The inorganic salt was filtered off. The solvent was removed from the filtrate by evaporation. The final product was obtained by distillation under vacuum using a Kugelrohr. The yield was 8.2 g. The boiling point and NMR data were

Compound 6: tris(hexafluoro-2-phenylisopropyl) borate, boiling point: 125 to $130^\circ\text{C}/0.05\tau$; NMR (acetone- d_6): 7.65 (s). (unit: ppm).

General procedure for preparation of fluorinated triaryl borate compounds: compounds 8, 9, 10, 11, 12, and 13.—An equivalent amount of 1 M boron trichloride solution in hexane (70 mL) was added to a solution of 0.21 mol of the corresponding fluorinated phenol compound in 250 mL of anhydrous methylene chloride at -78°C , over a period of 20 min. The reaction mixture was warmed to room temperature and stirred for 12 h. After removal of the solvent, the final product was further purified through either sublimation or distillation under vacuum. The respective boiling points and NMR data were

Compound 9: tris(2,4-difluoroethyl) borate, boiling point: 95 to $100^\circ\text{C}/0.06\tau$; NMR (in CDCl_3): 6.6–7.2(m). (unit: ppm).

Compound 10: tris(2,3,5,6-tetrafluorobutyl) borate, boiling point: 120 to $125^\circ\text{C}/0.05\tau$; NMR (in CDCl_3): 6.5–7.2(m). (unit: ppm).

Compound 11: tris(pentafluorophenyl) borate, boiling point: $115^\circ\text{C}/0.15\tau$; NMR (in CDCl_3): 6.5–7.2(m). (unit: ppm.)

Compound 12: tris(3-trifluoromethylphenyl) borate, boiling point: 128 to $130^\circ\text{C}/0.1\tau$; NMR (in CDCl_3): 7.4(s). (unit: ppm).

Compound 13: tris(3,5-bis(trifluoromethyl)phenyl) borate, boiling point: $120^\circ\text{C}/0.09\tau$; NMR (in CDCl_3): 7.6(s, 2H), 7.7(s, 1H). (unit: ppm).

Procedure for preparation of tris(pentafluorophenyl) borane (compound 14).—7.2 g of magnesium turnings was suspended in 300 mL of anhydrous ether. 74.2 g of bromopentafluorobenzene was added dropwise to the suspension. The addition was controlled at a rate which kept the ether refluxing gently. After the addition was completed, the reaction mixture was refluxed for another 1.5 h. 14.2 g (12.7 mL) of borontrifluoride diethyl etherate was added dropwise to the black solution at room temperature. Then 350 mL of toluene was added to the mixture. The mixture was refluxed for 1 h. After the ether was distilled off, the remaining toluene solution was heated to 100°C and kept at 100°C for 18 h. The inorganic salt was filtered off from the mixture. The straw-colored solid crude product was obtained after removing the toluene solvent by evaporation. The crude product was purified by sublimation at 100 to $110^\circ\text{C}/0.05\tau$. The pure product, a colorless crystalline substance, was obtained through recrystallization from hexane. The yield was 50%. The melting temperature was 125 to 127°C .

Conductivity measurements.—Conductivity measurements were made at 25°C using a Hewlett-Packard 4129A impedance analyzer in the frequency range from 5 Hz to 10 MHz. Cells with Pt electrodes were used for the measurements. The cell constants were calibrated by measuring a 0.05 N KCl standard aqueous solution. The ionic conductivities of DME solutions containing various boron-based compounds and lithium salts were compared with the conductivities of DME solutions containing lithium salts only.

NEXAFS measurements.—NEXAFS measurements were made at the Cl K edge on LiCl/DME electrolytes with and without boron based anion receptors. The measurements were done at Beam Line X19A of the National Synchrotron Light Source. The data were collected as fluorescence excitation spectra using a large solid angle ionization chamber as the fluorescence detector. The solutions after being used in conductivity measurements were poured into cells with thin Mylar windows for NEXAFS studies. Similar measurements were done at the iodine L_{I} edge for Li/DME solutions.

Results and Discussion

Table I lists the ionic conductivities of 0.2 M lithium salts in pure DME solutions, as well as the data for same solutions with 0.2 M additions of the various boron-based compounds. The ionic conductivities of all the lithium salts in pure DME solutions are very low. The conductivity depends on the anion and increases in the order: $\text{Cl}^- < (\text{C}_2\text{F}_5\text{COO})^- < (\text{CF}_3\text{COO})^- < \text{I}^-$. The LiF salt is insoluble in pure DME. Addition of compound 1, the unfluorinated alkyl borate,

Table I. Ionic conductivity of 0.2 M boron compound + 0.2 M lithium salt in DME.

Boron compounds	Lithium salts				
	CF ₃ COOLi	C ₂ F ₅ COOLi	LiF Conductivity (S/cm)	LiCl	LiI
Salt only	3.3×10^{-5}	2.1×10^{-5}		5.0×10^{-6}	7.3×10^{-4}
1. (CH ₃ O) ₃ B	2.3×10^{-5}	1.7×10^{-5}		5.0×10^{-6}	8.3×10^{-4}
2. (CF ₃ CH ₂ O) ₃ B	1.9×10^{-4}	7.0×10^{-5}	1.6×10^{-5}	5.4×10^{-5}	1.6×10^{-3}
3. (C ₂ F ₅ CH ₂ O) ₃ B	2.6×10^{-4}	1.1×10^{-4}	1.5×10^{-5}	8.4×10^{-5}	1.8×10^{-3}
4. [(CF ₃) ₂ CHO] ₃ B	3.9×10^{-3}	4.6×10^{-3}	1.3×10^{-5}	3.8×10^{-3}	4.1×10^{-3}
5. [(CF ₃) ₃ CO] ₃ B	3.8×10^{-3}	3.8×10^{-3}		3.0×10^{-3}	2.8×10^{-3}
6. [(CF ₃) ₂ C(C ₆ H ₅)O] ₃ B	3.8×10^{-5}	2.0×10^{-5}		4.6×10^{-6}	7.0×10^{-4}
7. (C ₆ H ₅ O) ₃ B	1.3×10^{-4}	6.3×10^{-5}		1.2×10^{-5}	1.2×10^{-3}
8. (FC ₆ H ₄ O) ₃ B	6.2×10^{-4}	4.0×10^{-4}	1.7×10^{-5}	6.2×10^{-5}	1.8×10^{-3}
9. (F ₂ C ₆ H ₃ O) ₃ B	1.5×10^{-3}	1.4×10^{-3}	1.7×10^{-4}	2.7×10^{-4}	2.1×10^{-3}
10. (F ₃ C ₆ HO) ₃ B	2.3×10^{-3}	2.0×10^{-3}	8.8×10^{-4}	1.7×10^{-3}	2.2×10^{-3}
11. (C ₆ F ₅ O) ₃ B	3.3×10^{-3}	3.7×10^{-3}	2.9×10^{-3}	3.0×10^{-3}	3.2×10^{-3}
12. (CF ₃ C ₆ H ₄ O) ₃ B	2.2×10^{-3}	2.3×10^{-3}		1.1×10^{-3}	2.6×10^{-3}
13. [(CF ₃) ₂ C ₆ H ₃ O] ₃ B	2.8×10^{-3}	3.4×10^{-3}		3.0×10^{-3}	3.4×10^{-3}
14. (C ₆ F ₅) ₃ B	3.2×10^{-3}	3.8×10^{-3}	1.7×10^{-3}	2.6×10^{-3}	2.2×10^{-3}

(CH₃O)₃B, had a minimal effect on conductivity. However, upon addition of compound 2, (CF₃CH₂O)₃B, which has one CF₃ electron withdrawing group attached to the end of the alkyl chain, the solubility of the salts and the conductivity of the solutions were both increased significantly. With compound 3, (C₂F₅CH₂O)₃B, which has a longer CF₃CF₂CF₂ electron withdrawing group on the alkyl chain, the conductivities increased even more. The most significant increase of conductivity of the lithium salt solutions was obtained when compound 4, [(CF₃)₂CHO]₃B, the branched tris(2H-hexafluoroisopropyl)borate, was added. Comparing the structure of compound 3 and compound 4, it is clear that the better conductivity enhancement is due to the two branched CF₃ groups which have much stronger electron withdrawing effect than the linear CF₃CF₂CF₂ group. The ionic conductivity reached to 3.9×10^{-3} S/cm for CF₃COOLi, 4.6×10^{-3} S/cm for C₂F₅COOLi, 3.8×10^{-3} S/cm for LiCl, and 4.1×10^{-3} S/cm for LiI. When compound 4 was used as an additive, the solubility and conductivity of LiCl in DME was greatly improved. However, the solubility of LiF in DME was not improved significantly by using compound 4 as an additive, and the conductivity was 1.3×10^{-5} S/cm. Encouraged by the results with compound 4, we synthesized compounds 5 [(CF₃)₃CO]₃B, and 6 [(CF₃)₂C(C₆H₅)O]₃B. The hope was for further conductivity enhancement by the addition of an extra CF₃ electron withdrawing group, in compound 5 and a phenyl group in compound 6. However, the conductivities of solutions containing compounds 5 and 6 were lower than those containing compound 4. The larger decrease in conductivity of solutions containing compound 6 as compared to compound 5 indicates that the decrease may be due to a steric hindrance effect, since the phenyl group in compound 6 is much larger than the CF₃ group in compound 5.

Compounds 7, (C₆H₅O)₃B; 8, (FC₆H₄O)₃B; 9, (F₂C₆H₃O)₃B; 10, (F₃C₆HO)₃B; and 11, (C₆F₅O)₃B are triaryl borate compounds with a different number of H atoms substituted by fluorine. When the unfluorinated compound 7 was used as an additive, the conductivities of the solutions of all the lithium salts increased to about the same level as found with compound 3. When compounds 8, 9, 10, and 11 were added, the ionic conductivities increased with increasing number of fluorine atom substitutions. For example, when these compounds were added to LiF/DME solutions, the conductivity increased from 1.7×10^{-5} S/cm for compound 8, to 1.7×10^{-4} S/cm for compound 9, to 8.8×10^{-4} S/cm for compound 10, and to 2.9×10^{-3} S/cm for compound 11. The conductivities of solutions of all the other salts followed a similar trend.

Compounds 12 and 13 were synthesized to examine the effects of different electron withdrawing groups on the aryl groups. When CF₃ was used in place of F to substitute the H atoms, the enhancement of conductivity was slightly better.

The addition of the fully fluorinated triaryl borane, compound 14, (C₆F₅)₃B, also yielded a very good enhancement in the conductivity.

The effects of the concentration of additives and salts on the conductivity were also studied. The results are listed in Table II. The three additives, compounds, 4, 11, and 14, showing the best conductivity enhancement at 0.2 M concentration, were further studied at higher concentrations (0.2, 0.3, 0.4, 0.5, 0.8, 1.0, and 1.2 M). Three lithium salts, CF₃COOLi, C₂F₅COOLi, and LiF were used in this study. Because the solubility of the salts, especially LiF, is related to the concentration of the additives, the concentration of the salts was chosen to be the same as the concentration of the additive in each solution. For all three additives studied, a maximum in conductivity was reached at 0.8 M for both CF₃COOLi and C₂F₅COOLi salts. Since the solubility of LiF in DME is too low when compound 4 was used as an additive, no conductivity measurements were

Table II. Ionic conductivity of lithium salts and boron additives with various concentrations in DME.

4: [(CF ₃) ₂ CHO] ₃ B			
Concentration of salts and additives	Lithium salts		
	CF ₃ COOLi Conductivity	C ₂ F ₅ COOLi (S/cm)	LiF
0.2 M	3.9×10^{-3}	4.6×10^{-3}	
0.3 M	5.5×10^{-3}	6.3×10^{-3}	
0.4 M	6.6×10^{-3}	7.7×10^{-3}	
0.5 M	7.9×10^{-3}	8.5×10^{-3}	
0.8 M	8.9×10^{-3}	9.5×10^{-3}	
1.0 M	8.9×10^{-3}	9.2×10^{-3}	
1.2 M	8.2×10^{-3}	8.6×10^{-3}	
11: (C ₆ F ₅ O) ₃ B			
Concentration of salts and additives	Lithium salts		
	CF ₃ COOLi Conductivity	C ₂ F ₅ COOLi (S/cm)	LiF
0.2 M	3.3×10^{-3}	3.2×10^{-3}	4.1×10^{-3}
0.3 M	3.7×10^{-3}	4.4×10^{-3}	4.1×10^{-3}
0.4 M	4.5×10^{-3}	5.5×10^{-3}	5.6×10^{-3}
0.5 M	5.9×10^{-3}	6.1×10^{-3}	6.0×10^{-3}
0.8 M	5.6×10^{-3}	6.4×10^{-3}	
1.0 M	5.2×10^{-3}	5.8×10^{-3}	6.8×10^{-3}
1.2 M	4.4×10^{-3}	5.4×10^{-3}	
14: (C ₆ F ₅) ₃ B			
Concentration of salts and additives	Lithium salts		
	CF ₃ COOLi Conductivity	C ₂ F ₅ COOLi (S/cm)	LiF
0.2 M	3.2×10^{-3}	3.0×10^{-3}	1.7×10^{-3}
0.3 M	5.1×10^{-3}	4.2×10^{-3}	3.5×10^{-3}
0.4 M	6.3×10^{-3}	4.9×10^{-3}	4.7×10^{-3}
0.5 M	6.8×10^{-3}	5.4×10^{-3}	6.4×10^{-3}
0.8 M	6.7×10^{-3}	5.9×10^{-3}	6.6×10^{-3}
1.0 M	6.0×10^{-3}	5.4×10^{-3}	6.2×10^{-3}
1.2 M	5.4×10^{-3}	4.4×10^{-3}	

Concentrations of salts and additives are the same in all solutions.

made. In LiF solutions, a maximum in conductivity was reached at 1.0 M for compound 11 and at 0.8 M for compound 14. For all of the eight salt/additive combinations studied, at least one concentration reached a conductivity of 6×10^{-3} S/cm, an adequate conductivity for lithium ion batteries, providing other properties such as thermal and electrochemical stability are satisfactory. The results of electrochemical stability studies on these solutions will be published later.

Since both the anion and cation participate in ionic conduction, conductivity studies alone are insufficient evidence for anion complexation by these boron-based compounds. Ideally, this should be corroborated by X-ray diffraction studies on single-crystal samples. However, the complex of interest is formed in solution, in a liquid state rather than a crystal state. Therefore, X-ray diffraction cannot be used to study this complexation. The spectroscopic technique used in this study is near edge X-ray absorption fine structure (NEXAFS). NEXAFS spectra at the chlorine K edge and iodine L_i edge were used to study the coordination symmetry of the Cl^- and I^- ions in solutions. In a NEXAFS experiment at the chlorine K edge, as the incident x-ray energy is scanned through the chlorine K edge, the ejected 1s photoelectrons can undergo bound-state transitions. Coordination symmetry can affect the energy levels of the final states. Therefore, the fine structure of the absorption spectrum near the edge will provide information about the coordination symmetry of the Cl^- anion. The principle is the same for iodine L_i edge. The difference is that the photoelectrons are excited from 2s states to 5p states for the iodine L_i edge.

Figure 2 shows how the boron-based additives with different electron withdrawing groups have a dramatic effect on fine structure of the absorption spectra. NEXAFS data at the chlorine K edge are given for DME solutions of (a) 0.2 M LiCl, (b) 0.2 M LiCl + 0.2 M compound 4, and (c) 0.2 M LiCl + 0.2 M compound 14. The "white line" peak, between 2820 and 2840 eV, above the edge, is due to dipole-

allowed transitions to final states of p symmetry. The structure of the "white line" is very sensitive to the coordination of the absorbing atom, Cl^- , in these spectra. This technique has been used to study the complexation between Cl^- and aza-based anion receptors,⁶ the coordination of potassium in a KI/polyethylene oxide (PEO) complex,⁷ and ion pairing of potassium salts in plasticizers.⁸ The so-called white line refers to the strong absorption peak above the edge energy which appeared as a white line in the days when photographic film was used for detection. The white line in curve (a) for LiCl in DME is a broad peak, with a small shoulder on the low-energy side. The shoulder is labeled as feature A, and the main peak as feature B. Similar structures were found for Cl^- in a free ion state, such as LiCl and KCl in dilute aqueous solutions. In such a symmetric environment, the 3p final states of the Cl^- are close to being degenerate, and the transition feature is a broad peak. White line peak splitting can occur if there is an asymmetric distribution of atoms (or molecules) surrounding the Cl^- with strong bonding. The more asymmetric the distribution is, the stronger the splitting effect will be. The asymmetric local field experienced by each of the degenerate 3p final states of Cl^- is different, and the degeneracy is lifted. This results in a splitting of the absorption peak. When the LiCl salt was dissolved in DME, the white line splitting is almost undetectable, since the distribution of the solvent molecules has a spherical symmetry surrounding the Cl^- ions and the interaction between solvent and Cl^- is weak. We classify these anions as uncomplexed anions. In curve (b), the white line splitting was observed when compound 4 was added into the solution. Feature A emerges as a peak. This gives strong evidence that Cl^- is indeed complexed with the boron atom in compound 4. In curve (c) for a solution containing LiCl and compound 14, the white line is completely split into two sharp peaks. This is a clear indication that the Cl^- forms a strong complex with compound 14 in LiCl/DME solution.

Conductivity data for salt solutions with compounds 8, 9, 10, and 11 indicate that the anion complexation effect increases as the number of fluorine atoms is increased in the fluorinated triaryl boron compounds. It is interesting to see whether a similar trend is observed in the white-line splitting. The NEXAFS spectra at the Cl K edge for DME solutions of LiCl and containing compounds 8, 9, 10, and 11 in DME are plotted in Fig. 3 as curves (a), (b), (c), and (d), respectively. In curve (a), for compound 8, with only one H being substituted by F, the spectrum is the same as curve (a) in Fig. 2 for LiCl in DME, and no white line splitting was observed. Even though the conductivity data have shown a significant effect of this compound, the NEXAFS data are not sensitive enough to show any difference in the spectra. With compound 9, where two H atoms are substituted by F atoms, a small splitting of the white line was observed in curve (b). With compound 10 where four H atoms are substituted by F, the white-line splitting was clearly observed in curve (c). The trend in these results is in good agreement with the conductivity data in Table I. However, with compound 11 where all five H atoms are substituted by F atoms, the degree of white-line splitting did not follow the trend of curves (a), (b), and (c). The degree of white-line splitting of curve (d) was actually less than curve (c). Several measurements showed that this was a reproducible effect. The conductivity data certainly indicate that the anion complexation effect of compound 11 is stronger than compound 10. The weaker white-line splitting of curve (d) in Fig. 3 may be due to a more symmetrical coordination by compound 11.

It would be very interesting to get spectroscopic evidence for complex formation in LiF solutions. Unfortunately, the energy of the fluorine K edge is too low for NEXAFS studies. However, the NEXAFS spectra for LiI solutions can be obtained at the iodine L_i edge. The spectra are plotted in Fig. 4 for the following DME solutions: (a) 0.2 M LiI, (b) 0.2 M LiI + 0.2 M compound 4, and (c) 0.2 M LiI + 0.2 M compound 14. The results are very similar to those in Fig. 2 for the Cl K edge. White-line split-

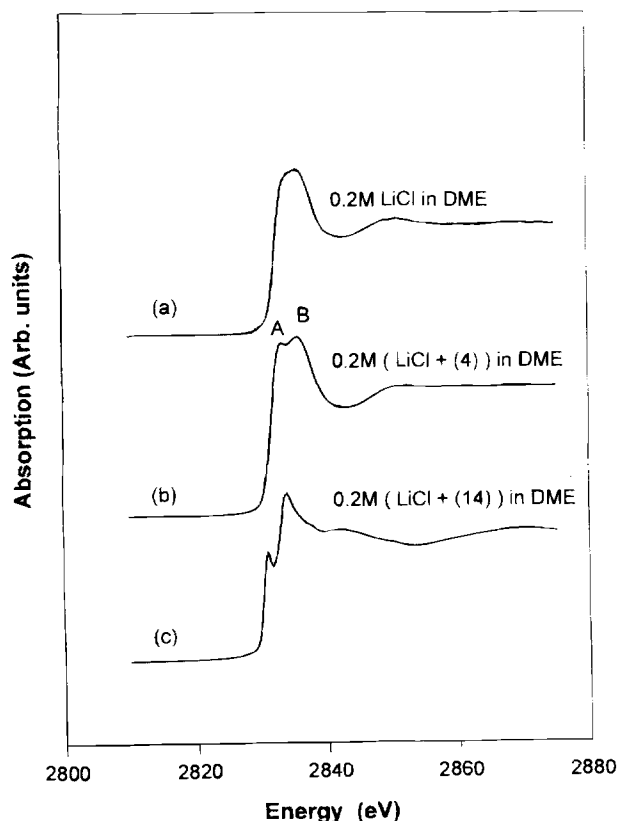


Fig. 2. Near-edge x-ray absorption spectra at the Cl K edge for DME solutions of: (a) 0.2 M LiCl salt, (b) 0.2 M [LiCl + compound (4)], and (c) 0.2 M [LiCl + compound (14)].

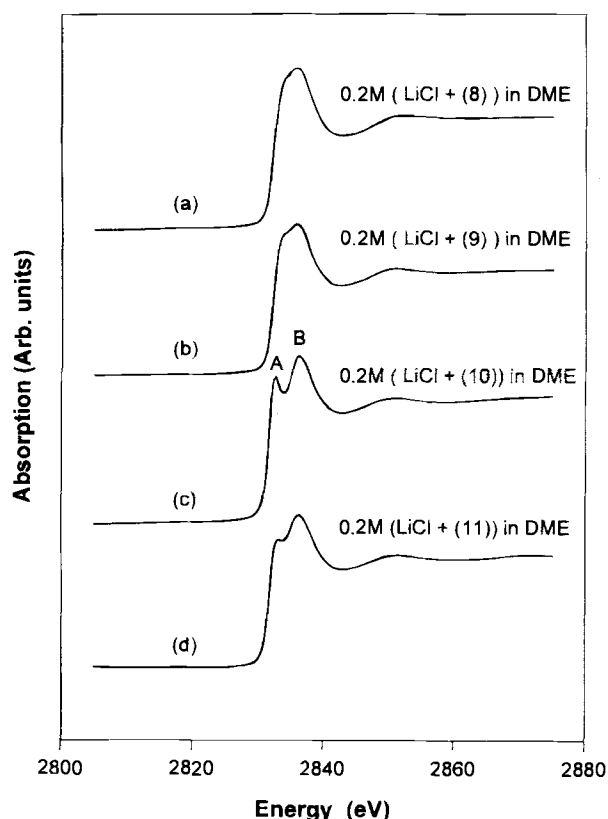


Fig. 3. Near-edge x-ray absorption spectra at the Cl K for DME solutions of: (a) 0.2 M [LiCl + compound (8)], (b) 0.2 M [LiCl + compound (9)], (c) 0.2 M [LiCl + compound (10)], and (d) 0.2 M [LiCl + compound (11)].

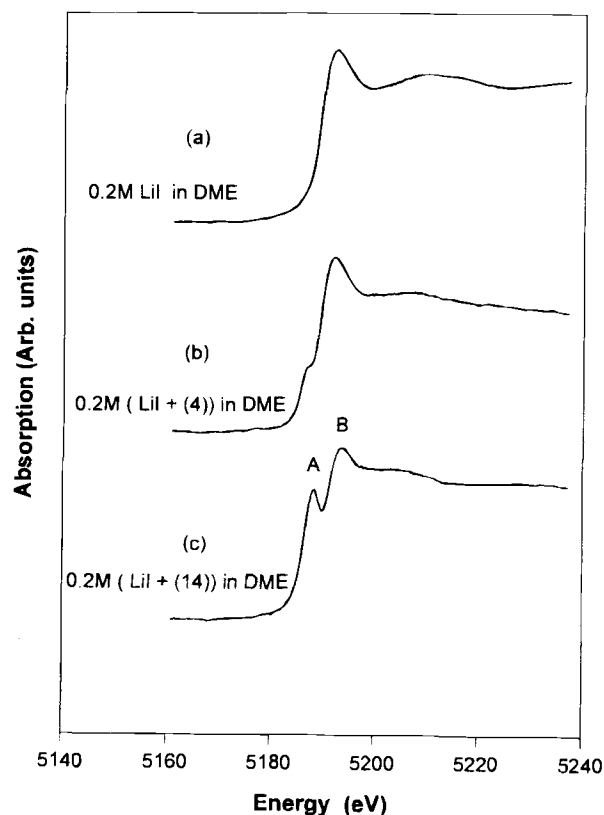


Fig. 4. Near-edge x-ray absorption spectra at the I L_1 edge for DME solutions of: (a) 0.2 M LiI, (b) 0.2 M [LiI + compound (4)], and (c) 0.2 M [LiI + compound (14)].

ting was observed in curve (b) for compound 4, and a much stronger white-line splitting was observed in curve (c) for compound 14.

The NEXAFS spectra at the iodine L_1 edge for DME solutions of LiI containing compounds 8, 9, 10, and 11 are plotted in Fig. 5 as curve (a), (b), (c), and (d), respectively. The results are similar to those in Fig. 3 for Cl K edge. The effect of white-line splitting is increased with increasing number of fluorine atoms in the boron compounds. There are two major differences between LiCl and LiI solutions. White-line splitting is observed in curve (a) for compound 8 in Fig. 5 whereas no significant splitting was observed in curve (a) of Fig. 3. The other difference is that the white-line splitting of curve (d) is stronger than curve (c) in Fig. 5, whereas the reverse is true in Fig. 3.

Conclusion

Ionic conductivity and NEXAFS studies show that a family of newly developed fluorinated alkyl and aryl borate and borane compounds form complexes with Cl^- and I^- anions and increase the ionic conductivity by increasing the concentration of free Li^+ cations, when used as additives in lithium salts/DME electrolytes. The degree of complexation and conductivity enhancement also depends on the structure of the boron compounds. The complexing agents have the ability to promote the dissolution of LiF, which is normally insoluble in organic solvents.

Acknowledgment

This work was supported by the U.S. Department of Energy Division of Materials Science of the Office of Basic Energy Sciences, under contract no. DE-AC02-98CH10886.

Manuscript received March 11, 1998.

Brookhaven National Laboratory assisted in meeting the publication costs of this article.

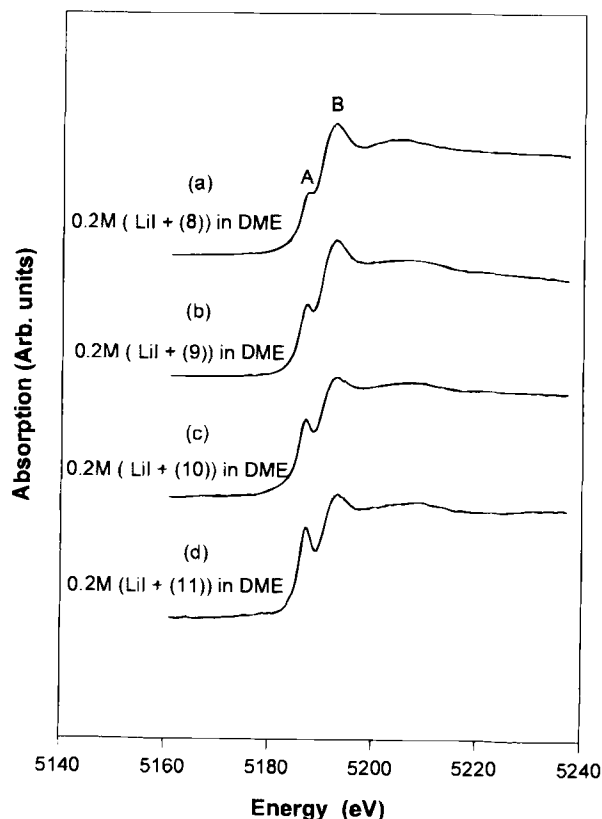


Fig. 5. Near-edge x-ray absorption spectra at the I L_1 edge for DME solutions of: (a) 0.2 M [LiI + compound (8)], (b) 0.2 M [LiI + compound (9)], (c) 0.2 M [LiI + compound (10)], and (d) 0.2 M [LiI + compound (11)].

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Comparison of Corrosion Resistances Between $\text{Fe}_2\text{O}_3\text{-TiO}_2$ Artificial Passivation Film and Passivation Film on Fe-Ti Alloy

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ABSTRACT

$\text{Fe}_2\text{O}_3\text{-TiO}_2$ double-oxide films were formed on Pt substrates by low-pressure metallorganic chemical vapor deposition using iron acetylacetonate and tetraisopropoxytitanium as precursors and O_2 as a reaction gas. Fe-Ti alloys were fabricated by an ion-beam sputter and an arc-melting process. The corrosion resistances of the $\text{Fe}_2\text{O}_3\text{-TiO}_2$ films and the Fe-Ti alloys were examined in acid solutions. The thinning rate of the $\text{Fe}_2\text{O}_3\text{-TiO}_2$ films in 5 M HCl decreased with an increase in the formation temperature. Likewise, the dissolution rate decreased with increasing titanium cationic fraction in the films. The films hardly dissolved at anodic potentials, while significant dissolution occurred at cathodic potentials in 1 M H_2SO_4 owing to the selective reduction of iron oxide components in the films. Sputter-deposited Fe-Ti alloy films containing more than 39 atom % Ti and arc-melted Fe-Ti alloys containing more than 57 atom % Ti showed high corrosion resistance when passivated anodically in 5 M HCl. The dissolution rates of the $\text{Fe}_2\text{O}_3\text{-TiO}_2$ films in 5 M HCl appear to be significantly smaller than those of the passive films on the Fe-Ti alloy films and the Fe-Ti alloys in the same solution.

Introduction

It is challenging to examine the real corrosion and electrochemical nature of passive films on practical alloys, because defects in the passive films, associated with crystal imperfections and nonmetallic inclusions in the alloy substrate, usually deteriorate the corrosion resistance of the films. One strategy for studying the real nature of the films involves separating the passive films from the alloy matrix. However, it is currently impossible to obtain such films in perfect condition. However, artificial passivation films, which simulate the composition, crystal structure, and thickness of passive films on corrosion-resistant alloys, are known as useful experimental materials for studying corrosion and the electrochemical nature of passive films on the alloys. For example, Sugimoto et al.¹⁻³ have produced $\text{Fe}_2\text{O}_3\text{-Cr}_2\text{O}_3$ and $\text{Fe}_2\text{O}_3\text{-Cr}_2\text{O}_3\text{-NiO}$ films, which simulate passive films on ferritic and austenitic stainless steels. The materials are prepared by metallorganic chemical vapor deposition (MOCVD) and their corrosion resistance examined as a function of the film composition. Similarly, Virtanen et al.⁴ and Schmuki et al.^{5,6} have formed Fe oxide and Cr oxide films by sputter-deposition and examined the cathodic reduction and transpassive dissolution properties of these oxides.

It is also possible to form artificial passive film that might be associated with unconventional corrosion-resistant alloys, such as $\text{Fe}_2\text{O}_3\text{-TiO}_2$ films. Sugimoto et al.^{7,8} synthesized the $\text{Fe}_2\text{O}_3\text{-TiO}_2$ double-oxide films by MOCVD using iron(III) acetylacetonate [$\text{Fe}(\text{O}_2\text{C}_5\text{H}_7)_3$] and tetraisopropoxytitanium [$\text{Ti}(\text{O}-i\text{-C}_3\text{H}_7)_4$] as precursors and reported the corrosion and the electrochemical behavior of the films in acid solutions. Their results showed the corrosion resistance in 5 M HCl increased with increasing Ti cationic fraction of the films⁷ and that no dissolution occurred at anodic potentials in 1 M H_2SO_4 .⁸ The photoelectrochemical behavior of the films was also examined in 0.5 M H_2SO_4 and it was found that there is a specific composition range

in which both the photoelectrical conversion efficiency and the corrosion resistance are extraordinarily high.⁹

There are no practical Fe-Ti alloys because equilibrium alloys in this system usually have dual-phase structures.¹⁰ Nonequilibrium alloys in this system, however, could have single-phase structure. If such alloys are obtained, $\text{Fe}_2\text{O}_3\text{-TiO}_2$ double-oxide films are expected on the alloys by spontaneous or anodic oxidation. In this case, comparison between the corrosion resistance of artificially synthesized $\text{Fe}_2\text{O}_3\text{-TiO}_2$ films and that of anodically formed $\text{Fe}_2\text{O}_3\text{-TiO}_2$ films is of interest, because the artificial $\text{Fe}_2\text{O}_3\text{-Cr}_2\text{O}_3$ films are reported to have higher corrosion resistance compared to anodic passivation films on Fe-Cr alloys.¹

The purpose of the present investigation is to make clear the detail of corrosion and electrochemical characteristics of the $\text{Fe}_2\text{O}_3\text{-TiO}_2$ artificial passivation films and to compare the corrosion resistance of such films with that of passive films formed on Fe-Ti alloys with homogeneous composition and structure. To realize this purpose, the $\text{Fe}_2\text{O}_3\text{-TiO}_2$ artificial passivation films were formed by the MOCVD at various temperatures and their corrosion resistances were examined by an immersion test in 5 M HCl and an electrochemical polarization test in 1 M H_2SO_4 . Nonequilibrium Fe-Ti alloys were prepared by ion-beam sputter deposition process and their corrosion resistances were also examined in the same environment. The corrosion resistance of passive films formed on dual-phase arc-melted alloys was also examined for reference.

Experimental

Preparation of specimens.—The $\text{Fe}_2\text{O}_3\text{-TiO}_2$ artificial passivation films were synthesized by low pressure (LP) MOCVD. Iron(III) acetylacetonate [$\text{Fe}(\text{O}_2\text{C}_5\text{H}_7)_3$, Dojindo Laboratories, 98%] and tetraisopropoxytitanium [$\text{Ti}(\text{O}-i\text{-C}_3\text{H}_7)_4$, High Purity Chemicals Laboratory, 99.999%] were used as vapor sources. N_2 (99.9998%) and O_2 (99.6%) were used as a carrier gas and a reaction gas, respectively. The LP-MOCVD apparatus and the deposition condition used in the present work were the same as those reported in

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